

Jongha Kim

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<https://jonghakim35.github.io/>

EDUCATION	Korea University M.S. & Ph.D. Integrated Student in Computer Science and Engineering Advised by Professor Hyunwoo J. Kim at MLV Lab GPA : 4.15 / 4.5 Korea University B.S. in Computer Science and Engineering B.S. in Statistics (Double Major) GPA : 4.22 / 4.5 (Major : 4.25 / 4.5)	Seoul, Republic of Korea Sep 2022 - Current Seoul, Republic of Korea Mar 2018 - Aug 2022
RESEARCH INTERESTS	My research focuses on multimodal foundation models for fine-grained understanding and reasoning over real-world visual data, particularly videos. I develop training, inference, and systems methods for grounded multimodal reasoning and efficient, scalable on-device deployment.	
WORK EXPERIENCES	Adobe Research • Research Scientist Intern at Video AI Lab • Mentor: Dr. Joon-Young Lee , Dr. Seoung Wug Oh , Dr. David Seunghyun Yoon • Designed an MLLM-based agentic workflow and a large-scale pseudo-labeling pipeline for referring video matting, improving supervision quality and scalability. MLV Lab • M.S. & Ph.D. Integrated Student at MLV Lab • Advisor: Prof. Hyunwoo J. Kim MLAI Lab • Undergraduate Research Intern at MLAI Lab • Advisor: Prof. Sung Ju Hwang Upstage AI • Teaching Assistant & Mentor of BoostCamp AI Tech 1-3rd course • Collaborated with Prof. Tae Hyun Oh VoyagerX • Intern at VoyagerX • Curated training data, optimized and deployed TensorFlow Lite (TFLite) on-device models for vFlat , a real-time mobile document scanner with 10M+ downloads on Google Play Store. SW Maestro • 10'th trainee at SW Maestro Program • Developed a computer vision model for estimating apple sugar content, leading to a registered Korean patent and a national big-data startup competition award.	San Jose, CA, USA May 2026 - Sep 2026 Seoul, Republic of Korea Sep 2022 - Current Seoul, Republic of Korea Jan 2021 - Jul 2022 Seongnam, Republic of Korea Dec 2020 - Jun 2022 Seoul, Republic of Korea Jul 2020 - Jan 2021 Seoul, Republic of Korea May 2019 - Nov 2019
PUBLICATION	[C7] Context Blindness in DPO: Mitigating Object Hallucination in MLLMs via Context-Calibrated Preference Optimization Byungoh Ko, Jinyoung Park, Jongha Kim, Jeehye Na, Jaewon Cho, Hyunwoo J. Kim European Conference on Computer Vision (<i>ECCV</i>), 2026 [C6] VideoSearch-R1: Iterative Video Retrieval and Reasoning via Soft Query Refinement Seohyun Lee*, Seoung Choi*, Dohwan Ko*, Jongha Kim, Hyunwoo J. Kim European Conference on Computer Vision (<i>ECCV</i>), 2026	

- [C5] **DocPrune: Efficient Document Question Answering via Background, Question, and Comprehension-aware Token Pruning**
Joonmyung Choi, Sanghyeok Lee, Jongha Kim, Sehyung Kim, Dohwan Ko, Jihyung Kil, Hyunwoo J. Kim
In collaboration with [Adobe](#) Research
IEEE/CVF Conference on Computer Vision and Pattern Recognition (*CVPR*), 2026
- [C4] **Relevance-aware Multi-context Contrastive Decoding for Retrieval-augmented Visual Question Answering**
Jongha Kim, Byungoh Ko, Jeehye Na, Jinsung Yoon, Hyunwoo J. Kim
In collaboration with [Google](#) Cloud AI
IEEE/CVF Winter Conference on Applications of Computer Vision (*WACV*), 2026
- [C3] **TabFlash: Efficient Table Understanding with Progressive Question Conditioning and Token Focusing**
Jongha Kim, Minseong Bae, Sanghyeok Lee, Jinsung Yoon, Hyunwoo J. Kim
In collaboration with [Google](#) Cloud AI
AAAI Conference on Artificial Intelligence (*AAAI*), 2026
- [C2] **VidChain: Chain-of-Tasks with Metric-based Direct Preference Optimization for Dense Video Captioning**
Ji Soo Lee*, Jongha Kim*, Jeehye Na, Jinyoung Park, Hyunwoo J. Kim
AAAI Conference on Artificial Intelligence (*AAAI*), 2025
- [C1] **Groupwise Query Specialization and Quality-Aware Multi-Assignment for Transformer-based Visual Relationship Detection**
Jongha Kim*, Jihwan Park*, Jinyoung Park*, Jinyoung Kim, Sehyung Kim, Hyunwoo J. Kim
IEEE/CVF Conference on Computer Vision and Pattern Recognition (*CVPR*), 2024
- [J1] **Improved Query Specialization for Transformer-based Visual Relationship Detection**
Jongha Kim, Jihwan Park, Jinyoung Park, Jinyoung Kim, Sehyung Kim, Hyunwoo J. Kim
Information Sciences, 2026
- [W2] **Concept Bottleneck with Visual Concept Filtering for Explainable Medical Image Classification**
Injae Kim*, Jongha Kim*, Joonmyung Choi, Hyunwoo J. Kim
MedAGI Workshop at International Conference on Medical Image Computing and Computer-Assisted Intervention (*MedAGI@MICCAI*), 2023 (**Oral Presentation**)
- [W1] **Object Detection in Aerial Images with Uncertainty-Aware Graph Network**
Jongha Kim, Jinheon Baek, Sung Ju Hwang
VOLI Workshop at European Conference on Computer Vision (*VOLI@ECCV*), 2022
- [U3] **SuperClip Pyramid for Video Temporal Grounding**
Sanghyeok Lee, Juyeon Ko, Joonmyung Choi, Jongha Kim, Dohwan Ko, Ji Soo Lee, Hyunwoo J. Kim
In collaboration with [KT](#) Gen AI Lab
Under Review
- [U2] **CEPO: Coordinator-Executor Prompt Optimization for Multi-Agent Document Understanding**
Jongha Kim*, Seohyun Lee*, Jonathan Minh Chung, Jinsung Yoon, Hyunwoo J. Kim
In collaboration with [Google](#) Cloud AI
Under Review
- [U1] **Reason in the Words You Speak: Idiolectal Paraphrasing Off-Policy Traces for Reasoning Distillation in VideoLLMs**
Ji Soo Lee, Jinyoung Park, Seohyun Lee, Jongha Kim, Joonmyung Choi, Jinsung Yoon, Hyunwoo J. Kim

In collaboration with Google Cloud AI
Under Review

(C: Conference, J: Journal, W: Workshop, U: Under Review, *: equal contribution)

PATENTS	Method and apparatus for dense video captioning based on chained task learning and metric loss for fine-grained understanding of video language model Hyunwoo Kim, Ji Soo Lee, <u>Jongha Kim</u> , Jeehye Na, Jinyoung Park Korea Patent No.10-2025-0164550. Information providing method and system for sharing fruit information including sugar content information measured through image vision processing Sanghoon Lee, Hyemin Song, <u>Jongha Kim</u> , Hyun Kim Korea Patent No.10-2020-0153010. Method for measuring sugar content of apple using image. Sanghoon Lee, Hyemin Song, <u>Jongha Kim</u> Korea Patent No.10-2242155, registered on Apr 14, 2021
HONORS & AWARDS	Travel Grant for CVPR 2024 Jun 2024 CVPR 2024 Graduate School Outstanding Freshman Scholarship Sep 2022 Korea University Dean's List Spring 2019 Korea University Semester High Honors Fall 2018/2019/2021, Spring 2019/2020/2021 Korea University 2020 Agrifood Public Big Data Startup Competition Aug 2020 Rural Development Administration National Scholarship for Science and Engineering Spring 2018 - Spring 2022 Korea Student Aid Foundation
ACADEMIC SERVICES	Reviewer of NeurIPS (2024–2026), CVPR (2024–2026), ICLR (2025–2026), AAAI (2025–2027), AISTATS (2025–2026), ICML (2025), WACV (2026–2027), MedAGI@MICCAI (2024–2026)
TALKS	Teaching Assistant Seoul, Republic of Korea Special Topics in Artificial Intelligence (AAA740), Korea University Fall 2024 Korea University & LG AI Workshop Seoul, Republic of Korea Korea University Feb 2023 Defining and solving problems in deep learning projects Virtual BoostCamp AI Tech 2nd course, Upstage AI Nov 2021
SKILLS	Language • Korean (<i>native</i>) • English (<i>fluent</i>), TEPS : 544/600 (<i>officially described as “Native Level of English Proficiency”</i>) Programming Skills • Python, PyTorch, Git, Bash (Most proficient) • TensorFlow, SQL, Node.js, React.js, Vue.js (Experienced)

Last Updated: Jul 7, 2026